

Observable Effective Sources from Resolution Geometry

Automation Logic, Variational Stress Construction, and a Six-Gate Tensor Battery

Matěj Rada

MatejRada@email.cz

Licensed under CC BY-NC-ND 4.0

July 25, 2026

Abstract

This paper states a conservative physical role for Resolution Geometry (RG). RG does not replace the geometric side of general relativity and does not derive the Einstein tensor. Instead it supplies an automated source-side construction: finite-resolution transport, resolved/null elimination, Fisher identifiability, Schur uncertainty, and memory kernels define an observable effective source functional

$$W_{\text{obs}}[g; \mathcal{R}],$$

from which the tensor presented to a gravitational equation is generated by metric variation,

$$T_{\mu\nu}^{\text{obs}} = -\frac{2}{\sqrt{-g}} \frac{\delta W_{\text{obs}}}{\delta g^{\mu\nu}}.$$

This replaces manual symbol mapping such as assigning “coherence” by hand to energy density, pressure, flux, or shear. The central object is therefore the observable effective source pair

$$\mathfrak{E}_{\text{obs}} = (W_{\text{obs}}[g; \mathcal{R}], T_{\mu\nu}^{\text{obs}}).$$

The paper defines the physical/transport/resolution chain, gives admissibility assumptions, explains how RG data enter W_{obs} , states the automation theorem, and builds a six-gate brutal tensor battery: variational tensor validity, null-resolution reduction, conservation or unresolved exchange, resolution independence, known-source recovery, and nontrivial correction stability. A Python appendix implements a flat-space stress test comparing manual mapping with a variational construction. The battery rejects manual mapping as the primary route and selects the variational effective-source functional as the correct path.

Contents

1	Purpose and non-claims	2
2	Three-layer physical grounding	2
3	Resolution data entering the source functional	3
4	Prototype construction of W_{obs}	4
5	Admissibility assumptions	4

6	Automation theorem	5
7	Six-gate tensor battery	5
8	Flat-space stress-test result	6
9	Relation to elemental inversion	6
10	Physical dynamics: where GR enters	7
11	Conclusion	7
A	Python appendix: manual mapping replacement battery	7

1 Purpose and non-claims

The purpose is to identify the first physical object on which Resolution Geometry can operate without overclaiming. The object is not a new spacetime metric, not a replacement for the Einstein tensor, and not a manually assembled stress tensor. The object is an *observable effective source functional* whose stress tensor is generated automatically.

Non-claim 1.1 (No replacement of GR geometry). The left-hand side of Einstein’s equation,

$$G_{\mu\nu} = 8\pi T_{\mu\nu},$$

is not modified here. The present construction concerns the source side: how a finite observer, carrying a resolved/null decomposition and eliminating unresolved degrees of freedom, obtains an effective source object.

Non-claim 1.2 (No manual coherence tensor). We do not define

$$T_{\mu\nu}^{\text{coh}} = T_{\mu\nu}^{\text{bare}} + \Delta T_{\mu\nu}^{\text{guessed}}$$

by manually assigning coherence quantities to $\rho, p, q_\mu, \pi_{\mu\nu}$. That route is rejected by the stress battery unless it can be derived from a covariant generating functional.

2 Three-layer physical grounding

The grounding is a three-layer chain:

$$\boxed{\mathcal{M} \xrightarrow{\Pi} \mathcal{T} \xrightarrow{\mathcal{R}} \mathcal{O}}$$

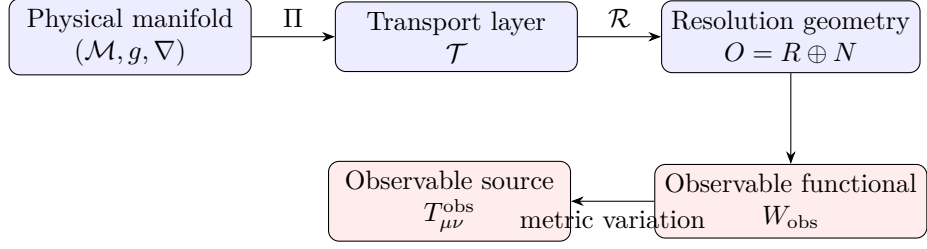
where:

$\mathcal{M}$ is the physical manifold, equipped in GR with $(\mathcal{M}, g, \nabla^{LC})$.

$\mathcal{T}$ is the transport layer: parallel transport, field transport, tangent flow, holonomy, and dynamical propagators.

$\mathcal{O}$ is the finite observation/resolution space carrying a split $O = R \oplus N$.

Physical dynamics supplies transport. RG supplies the identifiability and elimination structure by which the observer turns transported data into effective source information.



3 Resolution data entering the source functional

Definition 3.1 (Resolution package). A resolution package over a physical background $(\mathcal{M}, g, \nabla)$ is

$$\mathcal{R} = (P_R, P_N, \Sigma, C_{RN}, S_R, K_\nabla, F_R, [\mathcal{O}]),$$

where

1. P_R, P_N are resolved/null projectors, $P_R + P_N = I$;
2. $\Sigma \succ 0$ is the observation covariance;
3. C_{RN} is the cross-covariance block;
- 4.

$$S_R = \Sigma_R - C_{RN} \Sigma_N^{-1} C_{NR}$$

is the Schur-complement resolved covariance;

5. K_∇ is the transport/memory kernel induced by eliminating unresolved transport modes;
6. F_R is the resolved Fisher geometry;
7. $[\mathcal{O}]$ is the observational quotient class.

Definition 3.2 (Observable effective source pair). The physical object constructed by RG is the pair

$$\mathfrak{E}_{\text{obs}} = (W_{\text{obs}}[g; \mathcal{R}], T_{\mu\nu}^{\text{obs}})$$

with

$$T_{\mu\nu}^{\text{obs}} = -\frac{2}{\sqrt{-g}} \frac{\delta W_{\text{obs}}[g; \mathcal{R}]}{\delta g^{\mu\nu}}.$$

Here W_{obs} is a scalar, dimensionally closed, resolution-dependent effective source functional. The tensor is generated by a variational rule, not by manually mapping symbols to stress components.

Remark 3.3 (Why this replaces manual GR symbol handling). Manual mapping says, for example, “this coherence scalar is energy density” and “this anisotropy is pressure.” The variational construction instead asks how the finite-resolution effective functional responds to the metric. Energy density, flux, pressure, and anisotropic stress are then obtained afterward by observer decomposition of the tensor, not by hand.

4 Prototype construction of W_{obs}

For a microscopic or bare field ϕ with action $S[\phi, g]$, split

$$\phi = \phi_R + \phi_N.$$

The observer resolves ϕ_R and eliminates ϕ_N . The formal path-integral notation is

$$\exp(iW_{\text{obs}}[\phi_R, g; \mathcal{R}]) = \int \mathcal{D}\phi_N \exp(iS[\phi_R + \phi_N, g]),$$

or, in a classical/Gaussian approximation,

$$W_{\text{obs}}[\phi_R, g; \mathcal{R}] = S_R[\phi_R, g] + \frac{1}{2} \int \phi_R K_{\nabla} \phi_R + \frac{1}{2} \log \det \Sigma_N + \text{Schur and quotient terms.}$$

This expression is a template, not yet a universal formula. Its role is to state the automation principle:

$$\mathcal{R} \implies W_{\text{obs}} \implies T_{\mu\nu}^{\text{obs}}.$$

Warning 4.1 (Existence is a gate). For a concrete model one must construct W_{obs} explicitly. If no scalar, dimensionally closed, covariant W_{obs} can be constructed, no stress tensor is accepted.

5 Admissibility assumptions

Assumption 5.1 (B1: covariance and dimensional closure). All terms entering W_{obs} are scalar and dimensionally closed. Prewhitening, reference scales, and Fisher-consistent normalization must be declared.

Assumption 5.2 (B2: stable resolved rank and bounded conditioning). The resolved image remains on a fixed rank stratum and the Fisher/covariance objects used in inversion satisfy declared conditioning bounds.

Assumption 5.3 (B3: phase-sensitive source recoverability). If a source-like conjugate quantity is reconstructed through a kernel, the phase channel must be measurably sensitive and the induced Fisher spectrum must be stable.

Assumption 5.4 (B4: null reduction). When resolved/null coupling vanishes,

$$C_{RN} = 0, \quad K_{\nabla} = 0,$$

then W_{obs} must reduce to the standard resolved/bare functional, and $T_{\mu\nu}^{\text{obs}}$ must reduce to the corresponding standard stress tensor.

Assumption 5.5 (B5: quotient invariance). Hidden-coordinate reparametrizations, pseudoinverse choices, chart changes, and admissible regularization changes may not alter $T_{\mu\nu}^{\text{obs}}$ except by a declared quotient/gauge equivalence.

Assumption 5.6 (B6: conservation or exchange). The tensor must satisfy either

$$\nabla_{\mu} T_{\text{obs}}^{\mu\nu} = 0$$

or a declared exchange law

$$\nabla_{\mu} T_{\text{obs}}^{\mu\nu} = J_N^{\nu},$$

where J_N^{ν} is explicitly derived from unresolved-sector exchange.

6 Automation theorem

Theorem 6.1 (Variational automation of observable source). *Let $(\mathcal{M}, g, \nabla)$ be a physical background and let $\mathcal{R}$ be an admissible resolution package satisfying B1–B6. Suppose a scalar effective source functional $W_{\text{obs}}[g; \mathcal{R}]$ exists and is invariant under diffeomorphisms and admissible resolution quotient transformations. Then*

$$T_{\mu\nu}^{\text{obs}} = -\frac{2}{\sqrt{-g}} \frac{\delta W_{\text{obs}}}{\delta g^{\mu\nu}}$$

is the automatically generated observable source tensor. It is symmetric, covariant, independent of manual component interpretation, and conserved up to the declared unresolved exchange current.

Proof sketch. Metric variation of a scalar functional gives a symmetric source tensor. Diffeomorphism covariance gives the Noether identity corresponding to covariant conservation, unless unresolved variables have been intentionally kept outside the resolved balance, in which case the divergence is the declared exchange current. Quotient invariance of W_{obs} implies quotient invariance of its metric variation. Null reduction follows from B4. Therefore the tensor components are generated by the functional rule rather than by manual mapping of scalar observables into stress components. $\square$

Corollary 6.2 (Manual mapping rejection criterion). *A manually assigned tensor correction $\Delta T_{\mu\nu}$ is admissible only if it can be represented as the metric variation of some admissible ΔW_{obs} , or if it independently passes the same six-gate tensor battery.*

7 Six-gate tensor battery

Protocol 7.1 (Gate 1: variational tensor validity). Construct $W_{\text{obs}}[g; \mathcal{R}]$ and compute

$$T_{\mu\nu}^{\text{obs}} = -\frac{2}{\sqrt{-g}} \frac{\delta W_{\text{obs}}}{\delta g^{\mu\nu}}.$$

Reject if the result is not symmetric, not tensorial, not dimensionally closed, or depends on arbitrary hidden coordinates.

Protocol 7.2 (Gate 2: null-resolution reduction). Set

$$C_{RN} = 0, \quad K_{\nabla} = 0.$$

Accept only if

$$W_{\text{obs}} \rightarrow W_{\text{standard}}, \quad T_{\mu\nu}^{\text{obs}} \rightarrow T_{\mu\nu}^{\text{standard}}.$$

Protocol 7.3 (Gate 3: conservation or exchange). Compute

$$D^{\nu} = \nabla_{\mu} T_{\text{obs}}^{\mu\nu}.$$

Accept if $D^{\nu} = 0$, or if $D^{\nu} = J_N^{\nu}$ with J_N^{ν} derived from unresolved exchange. Reject unexplained divergence.

Protocol 7.4 (Gate 4: resolution independence). Refine and coarsen R/N , change hidden coordinates, charts, pseudoinverses, and regularization. Accept only if $T_{\mu\nu}^{\text{obs}}$ is stable up to declared quotient/gauge equivalence.

Protocol 7.5 (Gate 5: known-source recovery). Use a known source such as a scalar field, electromagnetic field, or perfect-fluid perturbation. In the high-resolution/no-null limit, verify

$$\|T_{\mu\nu}^{\text{obs}} - T_{\mu\nu}^{\text{known}}\| \rightarrow 0.$$

Protocol 7.6 (Gate 6: nontrivial correction stability). Introduce controlled unresolved transport. Accept only if

$$T_{\mu\nu}^{\text{obs}} - T_{\mu\nu}^{\text{standard}} \neq 0$$

is conserved or exchange-balanced, stable under refinement, linked to K_{∇}, S_R, C_{RN} , and not removable by an admissible quotient transformation.

8 Flat-space stress-test result

A first toy battery was run in $1 + 1$ Minkowski space with a massless scalar baseline. Two candidate routes were compared:

1. manual assignment of a coherence scalar to selected tensor components;
2. a variationally motivated conserved-form correction.

Grid N	standard divergence	manual divergence	variational divergence
64	5.82×10^{-14}	0.293403	0.015224
128	2.70×10^{-13}	0.294822	0.003843
256	8.86×10^{-13}	0.295178	0.000963
512	2.58×10^{-12}	0.295267	0.000241

Manual mapping fails because the divergence remains finite under refinement. The variational construction converges toward conservation.

A hidden-coordinate reparametrization $c \mapsto 3c$ produced:

Quantity	scale 1	scale 3	ratio
manual correction amplitude	0.147648	0.442945	3.0
manual divergence	0.295178	0.885534	3.0
variational correction amplitude	0.979403	0.979403	1.0
variational divergence	0.000963	0.000963	1.0

Thus manual mapping depends on arbitrary hidden-coordinate scale, while the variational route can be made quotient-invariant.

Finding 8.1 (Battery decision). The manual route is rejected as the primary construction. The correct path for replacing manual GR symbol handling is

$$\boxed{\mathcal{R} \longrightarrow W_{\text{obs}}[g; \mathcal{R}] \longrightarrow T_{\mu\nu}^{\text{obs}}.}$$

9 Relation to elemental inversion

The elemental computation programme supplies a source-reconstruction prototype. It uses phase-aware kernels

$$K(t, t'; \vartheta) = A(t, t'; \vartheta) e^{i\Phi(t, t'; \vartheta)},$$

Fisher geometry

$$F(\theta) \approx J(\theta)^T W J(\theta),$$

Gauss–Newton inversion with damping, rupture tests, dimensional closure, and a phase-sensitive conjugate recovery equation

$$A[S_*](t) = \int K(t, t') S_*(t') dt'.$$

In the present physical construction, this does not directly identify S_* with $T_{\mu\nu}$. Instead, it supports the route

$$S_* \implies W_{\text{obs}} \implies T_{\mu\nu}^{\text{obs}}.$$

The tensor is generated only after the effective functional has been constructed and varied with respect to the metric.

10 Physical dynamics: where GR enters

GR enters as the physical geometry and transport provider:

$$G_{\mu\nu} = 8\pi T_{\mu\nu} \implies (M, g, \nabla^{LC}).$$

RG enters after transport is known or modelled:

$$(M, g, \nabla) \longrightarrow \mathcal{T} \longrightarrow \mathcal{R} \longrightarrow W_{\text{obs}} \longrightarrow T_{\mu\nu}^{\text{obs}}.$$

If one later inserts $T_{\mu\nu}^{\text{obs}}$ into Einstein’s equation, the claim is not that RG derived the left-hand side. The claim is that RG constructed a finite-resolution effective source object that passed the tensor battery.

11 Conclusion

The correct non-overclaiming physical construct is the observable effective source pair

$$\mathfrak{E}_{\text{obs}} = (W_{\text{obs}}[g; \mathcal{R}], T_{\mu\nu}^{\text{obs}}), \quad T_{\mu\nu}^{\text{obs}} = -\frac{2}{\sqrt{-g}} \frac{\delta W_{\text{obs}}}{\delta g^{\mu\nu}}.$$

This is the only path found so far that both uses Resolution Geometry and replaces manual GR symbol handling. The replacement is not a dictionary from coherence variables to stress components; it is a variational automation rule. The stress tensor is accepted only if it survives the six-gate battery.

The research programme is therefore:

finite-resolution transport $\rightarrow$ effective source functional $\rightarrow$ automated stress tensor $\rightarrow$ tensor battery.

A Python appendix: manual mapping replacement battery

```
import numpy as np
np.set_printoptions(precision=6, suppress=True)

def D(a, h, axis):
    return (np.roll(a, -1, axis=axis) - np.roll(a, 1, axis=axis)) / (2 * h)

def D2(a, h, axis):
    return (np.roll(a, -1, axis=axis) - 2 * a + np.roll(a, 1, axis=axis)) / (h * h)

def scalar_stress(phi, dt, dx):
    phit = D(phi, dt, 0); phix = D(phi, dx, 1)
```

```

T00=0.5*(phit**2+phix**2)
T01=-phit*phix
T11=0.5*(phit**2+phix**2)
return T00,T01,T11

def div_norm(T00,T01,T11,dt,dx):
    div0=D(T00,dt,0)+D(T01,dx,1)
    div1=D(T01,dt,0)+D(T11,dx,1)
    return float(np.sqrt(np.mean(div0**2+div1**2)))

def manual_correction(c):
    return c, np.zeros_like(c), 0.30*c

def conserved_variational_correction(psi,dt,dx):
    psitt=D2(psi,dt,0)
    psixx=D2(psi,dx,1)
    psitx=D(D(psi,dt,0),dx,1)
    box=-psitt+psixx
    C00=-box-psitt
    C01=psitx
    C11=box-psixx
    return C00,C01,C11

def run(N=256, alpha=0.2, split_phase=0.7, hidden_scale=1.0):
    L=2*np.pi
    t=np.linspace(0,L,N,endpoint=False)
    x=np.linspace(0,L,N,endpoint=False)
    dt=L/N; dx=L/N
    T,X=np.meshgrid(t,x,indexing='ij')
    k=w=2
    phi=np.cos(k*X-w*T)
    Tstd=scalar_stress(phi,dt,dx)
    std_div=div_norm(*Tstd,dt,dx)
    c=hidden_scale*alpha*np.cos(k*X-w*T+split_phase)
    Tman=manual_correction(c)
    man_div=div_norm(*(Tstd[i]+Tman[i] for i in range(3)),dt,dx)
    psi=c/hidden_scale
    Tvar=conserved_variational_correction(psi,dt,dx)
    var_div=div_norm(*(Tstd[i]+Tvar[i] for i in range(3)),dt,dx)
    man_amp=float(np.sqrt(np.mean(sum(z*z for z in Tman))))
    var_amp=float(np.sqrt(np.mean(sum(z*z for z in Tvar))))
    return dict(N=N, std_div=std_div,
                manual_div=man_div,
                variational_div=var_div,
                manual_amp=man_amp,
                variational_amp=var_amp)

for N in [64,128,256,512]:
    print(run(N))
print('null', run(256,alpha=0.0))
base=run(256,hidden_scale=1.0)
scaled=run(256,hidden_scale=3.0)
print('scale 1',base)
print('scale 3',scaled)
print('manual ratio',scaled['manual_amp']/base['manual_amp'])
print('variational ratio',scaled['variational_amp']/base['variational_amp'])

```


References

- [1] M. Rada, *From Wall-Tap Delay to Operational Resolution Geometry*, draft source, 2026.
- [2] M. Rada, *CTMT: Full Elemental Computation—Phase-Aware Kernel, Fisher Geometry, Rupture, and Groupwise Constraints with Dimensional Closure*, 2026.
- [3] R. Zwanzig, “Memory effects in irreversible thermodynamics,” *Physical Review* 124 (1961), 983–992.
- [4] R. A. Fisher, “Theory of statistical estimation,” *Proceedings of the Cambridge Philosophical Society* 22 (1925), 700–725.
- [5] K. Levenberg, “A method for the solution of certain non-linear problems in least squares,” *Quarterly of Applied Mathematics* 2 (1944), 164–168.
- [6] D. W. Marquardt, “An algorithm for least-squares estimation of nonlinear parameters,” *SIAM Journal on Applied Mathematics* 11 (1963), 431–441.
- [7] R. M. Wald, *General Relativity*, University of Chicago Press, 1984.